

S3 Class - Student Mark Analysis System

S3 Class Student Mark Analysis System

Create S3 Class

```
student &lt;- function(id, name, marks){  
  structure(list(id=id, name=name, marks=marks), class="Student")  
}
```

Total Marks

```
total &lt;- function(x){  
  sum(x$marks)  
}
```

Average Marks

```
average &lt;- function(x){  
  mean(x$marks)  
  
}
```

Grade Function

```
grade &lt;- function(x){  
  a &lt;- average(x)  
  if(a >= 90)  
    "A"  
  else if(a >= 75)  
    "B"  
  else if(a >= 60)  
    "C"  
  else if(a >= 50)  
    "D"  
  else  
    "F"  
}
```

Result Function

```
result &lt;- function(x){
```

```

list(
Total = total(x),
Average = average(x),
Grade = grade(x)
)
}

# Print Method
print.Student <- function(x){
cat("<div data-bbox=
```

```
df <- data.frame(Subject=subjects, Marks=marks)

print(df)

# Pass Status

cat("Passed :", all(marks >= 50), "\n")

# Rank Score

cat("Rank Score :", average(s1), "\n")
```

```
> # Display Student Details
> print(s1)
Student ID : 101
Student Name : Yashwitha
Marks : 88 92 81 75 90
Total : 426
Average : 85.2
Grade : B
>
> # Highest and Lowest Marks
> highest <- max(s1$marks)
> lowest <- min(s1$marks)
>
> cat("Highest Marks :", highest, "\n")
Highest Marks : 92
> cat("Lowest Marks :", lowest, "\n")
Lowest Marks : 75
>
> # Subject-wise Marks
> subjects <- c("R", "DS", "AI", "ML", "DBMS")
>
> df <- data.frame(Subject=subjects, Marks=marks)
>
> print(df)
  Subject Marks
1      R     88
2     DS     92
3     AI     81
4     ML     75
5    DBMS     90
>
> # Pass Status
> cat("Passed :", all(marks >= 50), "\n")
Passed : TRUE
>
> # Rank Score
> cat("Rank Score :", average(s1), "\n")
Rank Score : 85.2
```

S4 Class - Student Mark Analysis System

```
# Create S4 Class
```

```
setClass("Student",
  slots = list(
    id = "numeric",
    name = "character",
    marks = "numeric"
  ))
```

```

# Generic Function

setGeneric("report", function(object)

  standardGeneric("report"))

# Method

setMethod("report", "Student",

  function(object){

    total <- sum(object@marks)

    avg <- mean(object@marks)

    grade <- if(avg >= 90) "A"

    else if(avg >= 75) "B"

    else if(avg >= 60) "C"

    else "F"

    cat("ID :", object@id, "\n")

    cat("Name :", object@name, "\n")

    cat("Marks :", object@marks, "\n")

    cat("Total :", total, "\n")

    cat("Average :", avg, "\n")

    cat("Grade :", grade, "\n")

    cat("Highest :", max(object@marks), "\n")

    cat("Lowest :", min(object@marks), "\n")

  })

# Student Marks

m <- c(85,78,92,88,80)

# Create Object

s <- new("Student",

  id = 1,

  name = "Yashwitha",

  marks = m)

# Display Report

report(s)

# Data Frame

```

```

df <- data.frame(
  Subject = c("R","AI","ML","DBMS","OS"),
  Marks = m
)

print(df)

# Pass Status

cat("Pass :", all(m >= 50), "\n")

```

```

> # Display Report
> report(s)
ID : 1
Name : Yashwitha
Marks : 85 78 92 88 80
Total : 423
Average : 84.6
Grade : B
Highest : 92
Lowest : 78
>
> # Data Frame
> df <- data.frame(
+   Subject = c("R","AI","ML","DBMS","OS"),
+   Marks = m
+ )
>
> print(df)
  Subject Marks
1      R     85
2     AI     78
3     ML     92
4    DBMS     88
5     OS     80
>
> # Pass Status
> cat("Pass :", all(m >= 50), "\n")
Pass : TRUE

```

Reference Class - Student Mark Analysis System

```

StudentRef <- setRefClass(
  "StudentRef",
  fields = list(
    id = "numeric",
    name = "character",
    marks = "numeric"
  )
)

```

```

),
methods = list(
  total = function() sum(marks),
  average = function() mean(marks),
  grade = function(){
    a <- average()
    if(a >= 90) "A"
    else if(a >= 75) "B"
    else if(a >= 60) "C"
    else "F"
  },
  display = function(){
    cat("ID:", id, "\n")
    cat("Name:", name, "\n")
    cat("Marks:", marks, "\n")
    cat("Total:", total(), "\n")
    cat("Average:", average(), "\n")
    cat("Grade:", grade(), "\n")
  }
)
)
obj <- StudentRef$new(
  id = 11,
  name = "Yashwitha",
  marks = c(91,82,77,88,95)
)
obj$display()

```

```
> obj <- StudentRef$new(  
+   id = 11,  
+   name = "Yashwitha",  
+   marks = c(91,82,77,88,95)  
+ )  
>  
> obj$display()  
ID: 11  
Name: Yashwitha  
Marks: 91 82 77 88 95  
Total: 433  
Average: 86.6  
Grade: B
```